

Q1

Explain the layered architecture of networks. Compare the OSI Reference Model (all 7 layers) and TCP/IP architecture (4 layers). Discuss the functions of each layer. What are the advantages and disadvantages of wired vs wireless networks?

Introduction: Computer networks use a layered architecture to break the complex problem of data communication into manageable, independent modules. Each layer performs specific functions and communicates only with the layers directly above and below it. The two dominant models are the OSI Reference Model and the TCP/IP Model.

Layered Architecture — Why? Layering reduces complexity, allows independent development of each layer, makes troubleshooting easier, and enables interoperability between different vendors' equipment. Changes in one layer don't affect others as long as interfaces remain the same.

OSI Reference Model (7 Layers):

Physical Layer (Layer 1): Transmits raw bits over a communication channel. Defines electrical, mechanical, and timing specifications. Deals with voltage levels, cable types, pin layouts, and data rates. Example: Ethernet cable, fiber optic.

Data Link Layer (Layer 2): Provides error-free transfer of frames between two directly connected nodes. Handles framing, MAC addressing, error detection (CRC), and flow control. Divided into LLC (Logical Link Control) and MAC sub-layers. Example: Ethernet, Wi-Fi (802.11).

Network Layer (Layer 3): Handles logical addressing (IP addresses) and routing — determining the best path for packets across multiple networks. Deals with packet fragmentation and reassembly. Example: IP, ICMP, routers.

Transport Layer (Layer 4): Provides end-to-end reliable (TCP) or unreliable (UDP) data delivery between processes. Handles segmentation, reassembly, flow control, congestion control, and error recovery. Port numbers identify applications.

Session Layer (Layer 5): Manages sessions (connections) between applications — establishing, maintaining, and terminating dialogues. Handles synchronization and checkpointing for long transfers.

Presentation Layer (Layer 6): Translates data between application format and network format. Handles encryption/decryption, compression, and character encoding (ASCII, Unicode). Example: SSL/TLS encryption.

Application Layer (Layer 7): Closest to the user. Provides network services to applications. Example protocols: HTTP, FTP, SMTP, DNS, SNMP.

TCP/IP Architecture (4 Layers):

Network Access Layer: Combines OSI Physical + Data Link. Handles hardware addressing (MAC), framing, and physical transmission. Protocols: Ethernet, ARP, PPP.

Internet Layer: Corresponds to OSI Network layer. Handles IP addressing and routing. Protocols: IP, ICMP, IGMP.

Transport Layer: Same as OSI Transport. TCP (reliable, connection-oriented) and UDP (unreliable, connectionless).

Application Layer: Combines OSI Session + Presentation + Application. HTTP, FTP, DNS, SMTP, SNMP.

OSI vs TCP/IP: OSI has 7 layers vs TCP/IP's 4. OSI is a theoretical reference model; TCP/IP evolved from working protocols. OSI's Session and Presentation layers are merged into TCP/IP's Application layer. TCP/IP is the model actually used on the Internet.

Wired vs Wireless Networks:

Wired (Ethernet, Fiber): High speed (up to 100 Gbps), low latency, highly secure (physical access needed to intercept), reliable, immune to interference. Disadvantage: expensive cabling, limited mobility, difficult to install in old buildings.

Wireless (Wi-Fi, 4G, 5G): Easy installation, supports mobility, flexible, cost-effective for coverage of large areas. Disadvantage: susceptible to interference (walls, microwaves, other devices), lower security (signals broadcast openly), higher latency, shared bandwidth among users, limited range.

Conclusion: The layered approach — whether OSI's 7-layer model or TCP/IP's practical 4-layer implementation — is fundamental to modern networking. Each layer's abstraction allows the Internet to function across billions of diverse devices. Understanding when to deploy wired versus wireless depends on the trade-off between mobility, cost, security, and performance requirements.

Q2

Explain different network topologies (Bus, Star, Ring, Mesh, Tree, Hybrid). Explain different transmission media (twisted pair, coaxial, fibre optic, wireless). Differentiate circuit switching vs packet switching.

Introduction: The physical and logical arrangement of a network — its topology — determines how devices communicate, how failures are handled, and how efficiently bandwidth is used. Physical layer transmission is subject to several impairments that degrade signal quality. The choice of transmission medium and switching technique further defines network performance.

Network Topologies:

Bus Topology: All devices share a single backbone cable. Simple and cheap to install. Disadvantage: a single cable failure brings down the entire network; performance degrades as more devices are added. Example: early Ethernet (10Base2).

Star Topology: All devices connect to a central hub or switch. Most common today. Failure of one device doesn't affect others; easy to add/remove devices; centralized management. Disadvantage: hub/switch failure disables entire network. Example: modern Ethernet LAN.

Ring Topology: Devices connected in a closed loop; data travels in one direction (or both in dual-ring). Deterministic access via token passing. Disadvantage: single break disrupts the ring. Example: Token Ring (IEEE 802.5), SONET.

Mesh Topology: Every device connects to every other device. Highly fault-tolerant and redundant. Disadvantage: expensive — $n(n-1)/2$ links needed for full mesh. Used in WANs and backbone networks. Example: Internet backbone.

Tree (Hierarchical) Topology: Combines star topologies in a hierarchy. Scalable and easy to manage. Used in large organizations with departments. Failure of a parent node isolates its subtree.

Hybrid Topology: Combination of two or more topologies. Most real networks are hybrid.

Transmission Impairments:

Attenuation: Loss of signal energy as it travels through the medium. Signal becomes weaker with distance. Solution: use amplifiers (analog) or repeaters (digital). Measured in decibels (dB).

Distortion: Change in signal shape, caused by different frequency components traveling at different speeds (dispersion). Particularly problematic in wired media. Solution: equalization filters.

Noise: Unwanted energy from external sources corrupting the signal. Types: thermal noise (random electron motion), intermodulation noise (signals mixing), crosstalk (signal from adjacent wire leaking in), impulse noise (sudden spikes from lightning or switching). Signal-to-Noise Ratio (SNR) measures how much signal dominates noise. Shannon's capacity theorem: $C = B \times \log_2(1 + \text{SNR})$ gives theoretical maximum channel capacity.

Transmission Media:

Twisted Pair: Two insulated copper wires twisted together to reduce crosstalk. UTP (Unshielded) is cheap and widely used for telephone and LAN (Cat5e: 1 Gbps, Cat6: 10 Gbps). STP (Shielded) adds metal shield for better noise immunity. Easy to install, low cost, limited bandwidth and distance.

Coaxial Cable: Central copper conductor surrounded by insulation, metallic shield, and outer jacket. Higher bandwidth and noise immunity than twisted pair. Used in cable TV, early Ethernet. Stiff, harder to install.

Fibre Optic: Transmits data as pulses of light through glass/plastic fibre. Extremely high bandwidth (Tbps), immune to electromagnetic interference, very low attenuation, secure (hard to tap). Expensive to install, requires skilled splicing. Single-mode fibre for long distances; multimode for shorter.

Wireless: Radio waves (Wi-Fi, Bluetooth, cellular), microwaves (satellite, point-to-point links), infrared (short range, line-of-sight). No physical cabling; supports mobility. Subject to interference and security concerns.

Circuit Switching vs Packet Switching:

Circuit Switching: A dedicated physical path is established before communication begins and held for the entire duration (e.g., traditional telephone networks). Guaranteed bandwidth, constant delay. Wasteful when channel is idle; call setup delay.

Packet Switching: Data is broken into packets that are individually routed through the network. No dedicated path; resources shared among many users. More efficient use of bandwidth, supports multiple simultaneous users, fault-tolerant. Variable delay (jitter). Used by the Internet. Packet switching advantages over circuit switching: better resource utilization (statistical multiplexing), no call setup delay, resilient to node failures (packets reroute), handles bursty traffic efficiently.

Q3

Explain error detection and correction in digital transmission: Parity Check, Checksum, CRC (Cyclic Redundancy Check) with polynomial method, and Hamming Code. What are flow control protocols? Explain Stop-and-Wait, Go-Back-N ARQ and Selective Repeat ARQ with sliding window protocol.

Introduction: In digital transmission, bits can be corrupted by noise, attenuation, or interference. Error control mechanisms detect and correct these errors to ensure reliable data delivery. Flow control prevents a fast sender from overwhelming a slow receiver. ARQ (Automatic Repeat reQuest) protocols combine both.

Error Detection Techniques:

Parity Check: A single parity bit is appended to each data unit. Even parity: total number of 1s (including parity bit) is even. Odd parity: total is odd. Simple, detects single-bit errors. Cannot detect even numbers of errors or correct errors. Example: data 1010001 → even parity bit = 1 → transmitted: 10100011.

Checksum: Data is divided into fixed-size segments. All segments are added together (1's complement arithmetic). The result is complemented and appended as the checksum. Receiver recomputes; if all-ones result → no error. Used in TCP, UDP, IP headers. Better than parity, detects most burst errors.

CRC — Cyclic Redundancy Check: Most powerful error detection. Treats the data as a large binary polynomial (message polynomial $M(x)$) and divides it by a generator polynomial $G(x)$. The remainder r is appended to the message. Receiver divides the received message by $G(x)$; if remainder is 0 → no error. Detects all single-bit errors, all double-bit errors, all odd numbers of errors, all burst errors of length $\leq$ degree of $G(x)$. Example: data = 1101011011, generator = 10011 → CRC computed by binary long division.

Error Correction — Hamming Code: Adds redundancy bits at power-of-2 positions. For m data bits, r parity bits needed where $2^r \geq m + r + 1$. Each parity bit covers specific bit positions. On receiving, recalculate all parity bits — the binary pattern of error positions (syndrome) directly gives the bit position that is wrong. Example: 4 data bits need 3 parity bits (positions 1, 2, 4). Can correct single-bit errors, detect double-bit errors.

Flow Control Protocols:

Stop-and-Wait ARQ: Sender transmits one frame, then waits for acknowledgement (ACK) before sending the next. Simple but highly inefficient — sender idle during propagation delay. Efficiency = $1/(1 + 2a)$ where a = propagation delay/transmission time.

Go-Back-N ARQ: Sender can have up to W (window size) unacknowledged frames in transit. If frame n is lost or corrupted, receiver discards frames $n, n+1, n+2, \dots$ and sender retransmits ALL from frame n onwards. Window size $\leq 2^n - 1$ (n = sequence number bits). Wastes bandwidth on retransmitting correct frames.

Selective Repeat ARQ: Like Go-Back-N but receiver buffers out-of-order frames. Only the specific corrupted frame is retransmitted. More efficient than Go-Back-N but requires larger receiver buffer. Window size $\leq 2^n - 1$.

Sliding Window Concept: Maintains two windows — sender window (frames that can be sent without ACK) and receiver window (frames that can be accepted). Window slides forward as ACKs are received. Allows pipelining — multiple frames in transit simultaneously, keeping the channel busy. **Conclusion:** Error detection techniques from simple parity to powerful CRC form the first defence against transmission errors, while Hamming code provides self-correction capability. ARQ protocols — from the simple but inefficient Stop-and-Wait to the bandwidth-efficient Selective Repeat — build reliable data links over unreliable physical channels using the sliding window principle.

Q4

Explain MAC (Medium Access Control) protocols: Pure ALOHA and Slotted ALOHA (with efficiency calculations), CSMA/CD — p-persistent, 1-persistent, non-persistent. Also explain functions of Switches and Routers and at which OSI layer they operate.

Introduction: In shared communication channels — such as a wireless network or early Ethernet bus — multiple devices compete for the medium simultaneously. Medium Access Control (MAC) protocols govern who can transmit when, preventing or recovering from collisions. This answer covers ALOHA, CSMA/CD variants, and the network devices that operate at the data link and network layers.

Pure ALOHA: Developed at the University of Hawaii (1970). Any station can transmit at any time. If a collision occurs (two stations transmit simultaneously), both wait a random backoff time and retransmit. Collision detected by lack of ACK within timeout. Maximum throughput: $S = G \times e^{-(2G)}$, peaks at $G = 0.5$, giving maximum efficiency of $1/(2e) \approx 18.4\%$. Very simple but highly inefficient — vulnerable time is $2 \times$ frame transmission time.

Slotted ALOHA: Time is divided into equal slots (= one frame transmission time). Stations can only begin transmission at the start of a slot, halving collision probability. Maximum throughput: $S = G \times e^{-(G)}$, peaks at $G = 1$, giving efficiency of $1/e \approx 36.8\%$ — double that of Pure ALOHA. Example: if $G = 1$ (one frame per slot on average), $P(\text{success}) = e^{-(1)} \approx 0.368$. Used in satellite communications and early GSM.

CSMA — Carrier Sense Multiple Access: Before transmitting, a station listens to the channel (carrier sense). If busy, it waits. If idle, it transmits. Reduces collisions but doesn't eliminate them (propagation delay means two stations may sense idle simultaneously).

1-Persistent CSMA: If channel busy, wait and transmit as soon as it becomes idle (probability 1). Greedy — high collision probability when multiple stations waiting.
Non-Persistent CSMA: If channel busy, wait a random time then sense again. Less greedy, better channel utilization, but longer delays when traffic is low.
p-Persistent CSMA: If channel idle, transmit with probability p ; defer to next slot with probability $(1-p)$. Balances between 1-persistent and non-persistent.

CSMA/CD (Collision Detection): Used in Ethernet (IEEE 802.3). Station senses the channel; if idle, transmits. While transmitting, continues to listen. If collision detected (signal level doubles), immediately stops, sends jam signal (32 bits) to ensure all stations detect the collision, then waits a random backoff time before retrying. Binary Exponential Backoff: after k -th collision, wait random time from $\{0, 1, ..., 2^k - 1\}$ slot times. Minimum frame size = $2 \times$ propagation delay $\times$ bandwidth (ensures frame still on wire when collision detected). CSMA/CA (Collision Avoidance) is used in wireless (Wi-Fi) since detecting collisions on wireless is impractical.

Switches vs Routers:

Switch (Layer 2 — Data Link): Connects devices within the same LAN. Forwards frames based on MAC addresses. Maintains a MAC address table (CAM table) — maps MAC addresses to physical ports. Learns by observing source MAC of incoming frames. Reduces collision domains (each port is its own domain). Supports full-duplex. Does not cross network boundaries. Example: 24-port office switch.
Router (Layer 3 — Network): Connects different networks. Forwards packets based on IP addresses. Maintains routing tables populated by static configuration or dynamic routing protocols (RIP, OSPF, BGP). Performs NAT (Network Address Translation), firewall functions, and QoS. Separates broadcast domains. Example: home broadband router, enterprise core router.

Conclusion: MAC protocols evolved from the simple but inefficient Pure ALOHA to CSMA/CD which dramatically improves channel utilization through carrier sensing and collision detection. Switches and routers operate at different OSI layers and serve complementary roles — switches organize traffic within a network segment efficiently using MAC addressing, while routers intelligently direct traffic between networks using IP addressing and routing algorithms.

Q5

Explain IP addressing — IPv4 classes (A, B, C, D, E), subnetting, CIDR. Compare IPv4 and IPv6 (why IPv6 is better). Explain DHCP and ARP. Describe private IP address ranges and why they are needed. Sketch the IP datagram format and explain each field.

Introduction: The Network layer is responsible for logical addressing and routing — determining how packets travel from source to destination across interconnected networks. IP (Internet Protocol) is the foundation of this layer, providing the addressing scheme. DHCP automates address assignment, while ARP resolves IP addresses to physical MAC addresses.

IPv4 Address Classes: IPv4 addresses are 32-bit numbers in dotted-decimal notation (e.g., 192.168.1.1). Divided into five classes:
Class A: First bit = 0. Range 1–126.x.x.x. Network: 8 bits, Host: 24 bits. 126 networks, ~16.7 million hosts each. For large organizations. Private: 10.x.x.x.
Class B: First bits = 10. Range 128–191.x.x.x. Network: 16 bits, Host: 16 bits. 16,384 networks, 65,534 hosts each. Medium organizations. Private: 172.16–31.x.x.
Class C: First bits = 110. Range 192–223.x.x.x. Network: 24 bits, Host: 8 bits. ~2 million networks, 254 hosts each. Small organizations. Private: 192.168.x.x.
Class D (224–239.x.x.x): Multicast. Class E (240–255.x.x.x): Reserved/experimental.
Special: 127.0.0.1 = loopback. 255.255.255.255 = broadcast.

Subnetting & CIDR: Subnetting divides a network into smaller subnets using a subnet mask. Example: 192.168.1.0/26 gives 4 subnets with 62 usable hosts each ($64 - 2$ for network and broadcast addresses). CIDR (Classless Inter-Domain Routing) uses prefix notation (/n) and eliminates rigid class boundaries, reducing routing table size through route aggregation (supernetting).

Private IP Addresses & NAT: Private ranges (10.x.x.x, 172.16–31.x.x, 192.168.x.x) are non-routable on the Internet, used within private networks. NAT (Network Address Translation) maps multiple private IPs to one public IP at the router, conserving IPv4 addresses. Advantage of subnetting: improves security, reduces broadcast domain size, better network management.

IPv4 Datagram Format Fields: Version (4 bits), Header Length (4 bits), Type of Service (8 bits, QoS), Total Length (16 bits), Identification + Flags + Fragment Offset (fragmentation), TTL (Time to Live — decremented each hop; prevents infinite loops), Protocol (TCP=6, UDP=17, ICMP=1), Header Checksum, Source IP, Destination IP, Options.

IPv6 — Why Better than IPv4: IPv6 uses 128-bit addresses ($2^{128} \approx 3.4 \times 10^{38}$ — effectively unlimited). Written as eight groups of hex digits. Key improvements: no address exhaustion, no NAT needed, built-in IPsec security, simpler fixed-size header (40 bytes, no checksum — left to transport layer), stateless address auto-configuration (SLAAC — no DHCP needed), better support for QoS and multicast, no broadcast (uses multicast instead), more efficient routing.

DHCP — Dynamic Host Configuration Protocol:

Automatically assigns IP addresses to hosts. Process (DORA):

1. *Discover:* Client broadcasts DHCP Discover (no IP yet).
2. *Offer:* DHCP server responds with IP address offer, subnet mask, gateway, DNS, lease time.
3. *Request:* Client broadcasts which offer it accepts.
4. *Acknowledge:* Server confirms the lease.

Lease is temporary — client must renew before expiry. DHCP removes manual IP configuration, prevents conflicts, and simplifies management.

ARP — Address Resolution Protocol: Bridges the gap between Layer 3 (IP) and Layer 2 (MAC). When a device wants to send a packet to an IP address on the same network, it doesn't know the MAC address. ARP broadcasts: "Who has IP 192.168.1.5? Tell 192.168.1.1." The device with that IP replies with its MAC address. The mapping is cached in the ARP table for future use. Gratuitous ARP: a device announces its own IP-to-MAC mapping (used to detect IP conflicts and update caches after NIC replacement).

Conclusion: IP addressing forms the logical foundation of Internet communication. IPv4's class-based scheme, while elegantly simple, has given way to CIDR and eventually IPv6 to meet the demands of a connected world. DHCP automates the critical task of address configuration, while ARP seamlessly bridges the logical IP world with the physical MAC addressing of the data link layer.

Q6

Explain routing protocols and algorithms: Distance Vector Routing (Bellman-Ford, count-to-infinity problem and 3 solutions), Link State Routing (Dijkstra's algorithm, OSPF). Explain ICMP — its purpose and message types. Explain unicast vs multicast routing.

Introduction: Routing is the process of selecting paths in a network along which to send data packets. Routing algorithms determine these paths. Static routing uses manually configured tables; dynamic routing uses protocols that automatically adapt to topology changes. This answer covers the two major dynamic routing paradigms and ICMP — the network layer's error and diagnostic protocol.

Distance Vector Routing (Bellman-Ford Algorithm): Each router maintains a distance vector table containing the best-known distance (hop count or metric) to every other router and which neighbour to go through. Periodically, each router sends its entire distance vector to its directly connected neighbours. Neighbours update their tables based on received vectors: $D(x, y) = \min \text{over all neighbours } v \{ \text{cost}(x, v) + D(v, y) \}$.

Convergence is gradual — good news (new short path) spreads fast, bad news (link failure) spreads slowly.

Count-to-Infinity Problem: When a link fails, routers may keep incrementing the distance to unreachable nodes, bouncing the metric up indefinitely. Example: A–B–C, if A–B link fails, B thinks it can reach A via C (C had A in its table from before), C thinks it can reach A via B → they loop, incrementing to infinity.

Solutions:

Split Horizon: A router never advertises a route back through the interface it learned it from. Prevents two-node loops.

Poison Reverse: Advertise failed routes with infinite metric (16 in RIP) back to the neighbour. More aggressive than split horizon.

Hold-Down Timers: When a route goes down, ignore any updates about that route for a fixed period — prevents premature re-addition of a bad route. Protocol: RIP (Routing Information Protocol) uses distance vector, max 15 hops, updates every 30 seconds.

Link State Routing (Dijkstra's Algorithm): Each router builds a complete map of the network topology. Process: Each router discovers its neighbours (Hello packets) and measures link costs.

Each router broadcasts Link State Advertisements (LSAs) to ALL routers in the network (flooding).

Every router builds identical network topology graph. Each router runs Dijkstra's Shortest Path First (SPF) algorithm on its map to compute best paths.

Dijkstra's: Start with source, tentatively assign infinite distance to all, zero to source. Repeatedly pick the unvisited node with smallest tentative distance, update its neighbours. Converges to shortest paths in $O((V+E) \log V)$. Advantages over distance vector: Faster convergence, no count-to-infinity, scales better with OSPF hierarchical areas. Disadvantage: more memory and CPU (each router stores full topology).

Protocol: OSPF (Open Shortest Path First) — link state, supports VLSM/CIDR, areas reduce flooding scope, uses cost = $10^8/\text{bandwidth}$ as metric.

ICMP — Internet Control Message Protocol: ICMP is used by network devices to send error messages and operational information. It rides inside IP packets (Protocol = 1). Key message types:

Echo Request/Reply (Type 8/0): Used by the ping command to test reachability. Sender sends Echo Request to destination IP; if reachable, destination replies with Echo Reply. Measures RTT.

Destination Unreachable (Type 3): Router or host cannot deliver packet. Codes indicate reason: Network unreachable (0), Host unreachable (1), Port unreachable (3), Fragmentation needed but DF bit set (4).

Time Exceeded (Type 11): TTL reached 0 before packet reached destination (Code 0 — used by traceroute). Each hop decrements TTL; when it hits 0 the router sends Time Exceeded back to source. Traceroute sends packets with TTL=1, 2, 3... to map each router hop.

Redirect (Type 5): Router tells host to use a better route.

Source Quench (Type 4, deprecated): Congestion control signal asking sender to slow down.

Unicast vs Multicast Routing: Unicast: one sender to one specific receiver. Standard IP forwarding. Multicast: one sender to a group of interested receivers. Uses Class D addresses (224.0.0.0–239.255.255.255). IGMP (Internet Group Management Protocol) lets hosts join multicast groups. PIM (Protocol Independent Multicast) routes multicast traffic efficiently, building a distribution tree. Much

Q7

Discuss the distinguishing features of TCP in detail: three-way handshake for connection setup (why not two-way?), reliable data transfer, flow control (sliding window), congestion control (slow start, congestion avoidance, fast retransmit), RTT estimation, and retransmissions. Also explain TCP connection management finite state machine.

Introduction: TCP (Transmission Control Protocol) is the cornerstone of reliable Internet communication. While IP provides best-effort packet delivery with no guarantees, TCP builds a reliable, ordered, connection-oriented service on top of IP. Understanding TCP's mechanisms is essential to understanding how the Internet actually achieves reliability for applications like web browsing, email, and file transfer.

Key Features of TCP: Connection-oriented (logical connection established before data transfer), full-duplex (simultaneous bidirectional data flow), byte-stream service (no message boundaries), reliable delivery (acknowledgements, retransmission), ordered delivery (sequence numbers), flow control (receiver window), congestion control (adaptive sending rate), error detection (16-bit checksum).

TCP Segment Header Fields: Source Port, Destination Port, Sequence Number (byte offset of first byte in segment), Acknowledgement Number (next expected byte), Header Length, Flags (SYN, ACK, FIN, RST, PSH, URG), Receive Window (flow control), Checksum, Urgent Pointer, Options (MSS, SACK, timestamps).

Three-Way Handshake (Connection Establishment):

1. **SYN:** Client sends segment with SYN flag set, ISN (Initial Sequence Number) = x. "I want to connect, my sequence starts at x."
2. **SYN-ACK:** Server responds with SYN + ACK. Server's ISN = y, ACK = x+1. "I accept, my sequence starts at y, I expect byte x+1 from you."
3. **ACK:** Client sends ACK = y+1. "I expect byte y+1 from you." Connection established.

Why Not Two-Way Handshake? A two-way handshake would only confirm one direction of communication. The server needs to know the client received its SYN-ACK and is ready to receive. Without the third ACK, the server doesn't know if its SYN-ACK arrived, and couldn't safely establish its receive window. Also prevents old duplicate connection requests from creating spurious connections.

Reliable Data Transfer: Every segment is acknowledged. If ACK not received within timeout (RTO), the segment is retransmitted. Cumulative ACKs acknowledge all bytes up to ACK number. TCP uses piggybacking — ACKs sent with data segments in the reverse direction.

RTT Estimation: Timeout (RTO) must adapt to network conditions. TCP measures SampleRTT for each segment. $EstimatedRTT = (1-\alpha) \times EstimatedRTT + \alpha \times SampleRTT$ ($\alpha = 0.125$). DevRTT measures variability. $RTO = EstimatedRTT + 4 \times DevRTT$. If timeout occurs, RTO is doubled (exponential backoff).

Flow Control — Receiver Window: Receiver advertises its available buffer space (rwnd) in every ACK. Sender limits unacknowledged data to $\min(cwnd, rwnd)$. Prevents fast sender overwhelming slow receiver. If $rwnd = 0$, sender stops but sends 1-byte probe segments to check when window reopens.

Congestion Control: *Slow Start:* cwnd starts at 1 MSS, doubles every RTT (exponential growth) until ssthresh is reached. *Congestion Avoidance:* Above ssthresh, cwnd increases by 1 MSS per RTT (linear growth). *Fast Retransmit:* On receiving 3 duplicate ACKs, immediately retransmit (don't wait for timeout). $ssthresh = cwnd/2$, $cwnd = ssthresh + 3$. *Fast Recovery:* After fast retransmit, continue in congestion avoidance (not slow start). *On timeout:* $ssthresh = cwnd/2$, $cwnd = 1$ MSS, restart slow start.

TCP Connection Termination — Four-Way Handshake:

Client FIN → Server ACK → (server continues sending if needed) → Server FIN → Client ACK. Client waits 2×MSL (Maximum Segment Lifetime = 2 min) in TIME-WAIT before fully closing — ensures final ACK reached server.

Conclusion: TCP's elegance lies in achieving reliable, ordered, flow-controlled, and congestion-controlled communication over the unreliable IP layer. The three-way handshake establishes mutual readiness; sequence numbers and ACKs ensure reliability; the sliding window provides flow control; and the AIMD (Additive Increase Multiplicative Decrease) congestion control algorithm ensures TCP flows share the network fairly without overwhelming it.

Q8

Explain network security: user authentication, data integrity, and cryptography (symmetric — DES, AES; asymmetric — RSA). Explain Firewalls — types and configurations (packet filtering, stateful, proxy). Explain DNS working and chain of DNS servers. Explain Quality of Service (QoS) parameters and how to improve QoS.

Network Security: Cryptography, Authentication, Firewalls, DNS & QoS

Introduction: As networks carry increasingly sensitive data, security becomes paramount. Network security encompasses mechanisms to ensure confidentiality (data not readable by unauthorised parties), integrity (data not altered in transit), authentication (verifying identity), and availability (services remain accessible). This answer covers cryptographic algorithms, authentication, firewalls, DNS, and Quality of Service.

Authentication: Authentication verifies that a communicating entity is who it claims to be. Methods: *Password-based:* Username + password. Simple but vulnerable to brute force, dictionary attacks, and eavesdropping if transmitted in plain text. *Challenge-Response:* Server sends a random challenge; client must respond with a value computed using a shared secret (e.g., HMAC-SHA256 of challenge+secret). Proves knowledge of secret without transmitting it. *Digital Certificates (PKI):* A Certificate Authority (CA) signs a certificate binding a public key to an identity. Used in HTTPS — browser verifies the server's certificate was signed by a trusted CA. *Multi-Factor Authentication (MFA):* Combines something you know (password), something you have (OTP on phone), and something you are (fingerprint).

Data Integrity: Achieved using cryptographic hash functions (MD5 — broken, SHA-1 — deprecated, SHA-256 — current standard). A hash is a fixed-size fingerprint of data. Any change to data produces a completely different hash (avalanche effect). To ensure integrity: sender computes hash, sends data + hash. Receiver recomputes hash; if it matches → data unmodified. HMAC (Hash-based Message Authentication Code) adds a secret key to the hash for combined integrity + authentication.

Symmetric Key Cryptography: Same key for encryption and decryption. Fast — suitable for bulk data. Key distribution is the main challenge. *DES (Data Encryption Standard):* 56-bit key, 16 rounds of substitution-permutation. Now broken — brute-forceable. *3DES:* Applies DES three times with different keys. 168-bit effective security. Slow. *AES (Advanced Encryption Standard):* 128/192/256-bit keys, 10/12/14 rounds. Current gold standard. Hardware-accelerated in modern CPUs. Used in HTTPS, disk encryption, VPNs.

Asymmetric Key Cryptography: Public key + private key pair. Encrypt with public key → only private key decrypts. Solves key distribution problem. *RSA:* Based on difficulty of factoring large primes. 2048-bit keys typical. Slow — used for key exchange and digital signatures, not bulk data. Example: HTTPS uses RSA to exchange a symmetric session key.

Firewalls:

Packet Filtering (Stateless): Examines each packet's source/destination IP, port, and protocol against ACL rules. Fast but cannot detect attacks spanning multiple packets. *Stateful Inspection:* Tracks TCP connection state. Allows inbound packets only for established outbound sessions. Much more secure than stateless.

Application-Layer Gateway (Proxy Firewall): Terminates the connection, inspects application content, then opens new connection if legitimate. Can filter URLs, commands, and content. Slowest but most thorough.

DNS — Domain Name System: DNS translates domain names to IP addresses. Hierarchy: Root servers → TLD servers (.com, .in) → Authoritative name servers. Resolution: Recursive resolver queries root → TLD → authoritative → returns IP to client. DNS records: A (IPv4), AAAA (IPv6), MX (mail), CNAME (alias), NS (name server). DNS uses UDP port 53 for queries, TCP port 53 for zone transfers.

Quality of Service (QoS): QoS guarantees performance levels for specific traffic types. Parameters: Bandwidth (throughput), Latency (delay), Jitter (variation in delay), Packet loss. Critical for real-time applications (VoIP, video conferencing) that cannot tolerate high latency or jitter. Techniques to improve QoS: *Traffic Classification:* Identify and mark packets by type (voice, video, data). *Prioritisation (QoS Queuing):* High-priority traffic (VoIP) served before bulk traffic (downloads). Priority Queuing, Weighted Fair Queuing (WFQ). *Traffic Shaping:* Leaky Bucket — enforces constant output rate regardless of burst input. Token Bucket — allows bursts up to a token bucket size while enforcing.